

D2233R2

2020 Fall Library Evolution Polls

Draft Proposal, 2020-10-22

Source:

[GitHub](#)

Issue Tracking:

[GitHub](#)

Project:

ISO/IEC JTC1/SC22/WG21 14882: Programming Language — C++

Audience:

WG21

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Introduction

This paper contains the decision polls that the C++ Library Evolution group will take in Fall of 2020.

Voting

These decision polls will be conducted electronically; see [\[P2195R0\]](#) for more information on electronic voting procedures.

All committee members may vote, but if you have not participated in the Library Evolution discussions of a poll, please choose to not vote.

How to Vote

[Vote here](#)

A documents.isocpp.org account is necessary to vote. If you have an isocpp.org/papers account, a documents.isocpp.org account has been created for you. Just go to documents.isocpp.org and reset your password. Email [Bryce Adelstein Lelbach](#) if you do not have an account or are having trouble accessing your account.

If you see "Form not found" when you go to the above link, you are not signed in to documents.isocpp.org. Go to documents.isocpp.org and sign in first.

You will not receive a confirmation email after voting. The Library Evolution chairs will send a confirmation to all voters before the close of the poll.

If you need to change how you voted or have any questions or issues about voting, please email [Bryce Adelstein Lelbach](#).

Voting Timeline

- 2020-10-13: Draft polls distributed by mailing list.
- 2020-10-22: Final discussion telecon on the polls. Start of the polling period.
- 2020-11-16: Chairs send a reminder to all voters of how they voted.
- 2020-11-19: End of the polling period.
- 2020-11-20: Chairs share results and summarize consensus and conclusions of the results.

Polls

Each poll consists of a statement; you vote on whether you support the statement. All polls will be 5-way polls; you'll vote either "strongly favor", "weakly favor", "neutral", "weakly against", or "strongly against". If you do not want to participate in a poll, select the "I do not want to participate in this poll" option. Those who do not participate in a poll will not be counted or recorded.

Each poll has a comment field. Please write a few sentences explaining the rationale for your vote.

Poll results and comments will be shared with Library Evolution after the end of the polling period, including attribution of your votes and comments.

If you want to discuss the polls or how you plan on voting, please start an email thread on the [Library Evolution mailing list](#).

Executor Polls

The following C++ Library Evolution polls relate to executors ([\[P0443R14\]](#)), a proposed set of abstractions for asynchronously creating and managing execution agents. These polls are based on the reports of the six breakout groups that reviewed various aspects of the executors proposal in the Summer of 2020 ([\[P2202R0\]](#), [\[P2203R0\]](#), [\[P2183R0\]](#), [\[P2205R0\]](#), [\[P2206R0\]](#), and [\[P2207R0\]](#)).

Poll 0

Remove implicit adaptation from [\[P0443R14\]](#) (Executors) by applying [\[P2235R0\]](#) to [\[P0443R14\]](#) which will:

- Make `schedule` only take schedulers.
- Make `execute` only take executors.
- Make sender and receiver operations like `connect` only take senders and receivers.
- Add explicit adaptation from executor to scheduler (`make_scheduler_from_executor`) but not vice versa.

Poll 1

Use one class for each individual trait instead of combined traits classes (`sender_traits`, etc) in [\[P0443R14\]](#) (Executors).

Poll 2

Remove `static_thread_pool` from [\[P0443R14\]](#) (Executors). It may be pursued in a follow-on proposal.

Poll 3

Remove `any_executor` from [\[P0443R14\]](#) (Executors). It may be pursued in a follow-on proposal.

Poll 4

Remove `any_executor::target` and `any_executor::target_type` from [\[P0443R14\]](#) (Executors).

Poll 5

[\[P0443R14\]](#) (Executors) is sufficiently mature that we should aim to ship it in C++23.

Other Polls

Poll 6

Send [\[P2212R1\]](#) (Relax Requirements for `time_point::clock`) to LWG for C++23, classified as an improvement of an existing feature ([\[P0592R4\]](#) bucket 2 item).

Poll 7

Send [\[P2166R1\]](#) (Prohibit `basic_string` and `basic_string_view` Construction from `nullptr`) to LWG for C++23, classified as an improvement of an existing feature ([\[P0592R4\]](#) bucket 2 item).

Poll 8

Send [\[P2161R0\]](#) (Remove Default Candidate Executor) to LWG for the Networking TS Working Draft, classified as a focus work item ([\[P0592R4\]](#) bucket 1 item).

References

Informative References

[P0443R14]

Jared Hoberock, Michael Garland, Chris Kohlhoff, Chris Mysisen, H. Carter Edwards, Gordon Brown, D. S. Hollman. [A Unified Executors Proposal for C++](#). 15 September 2020. URL: <https://wg21.link/p0443r14>

[P0592R4]

Ville Voutilainen. [To boldly suggest an overall plan for C++23](https://wg21.link/p0592r4). 25 November 2019. URL: <https://wg21.link/p0592r4>

[P2161R0]

Robert Leahy. [Remove Default Candidate Executor](https://wg21.link/p2161r0). 14 May 2020. URL: <https://wg21.link/p2161r0>

[P2166R1]

Yuriy Chernyshov. [A Proposal to Prohibit std::basic_string and std::basic_string_view construction from nullptr](https://wg21.link/p2166r1). 7 September 2020. URL: <https://wg21.link/p2166r1>

[P2183R0]

David Olsen, Ruslan Arutyunyan, Michael J. Voss, Michał Dominiak, Chris Kohlhoff, D.S. Hollman, Kirk Shoop, Inbal Levi. [Executors Review: Properties](https://wg21.link/p2183r0). 14 July 2020. URL: <https://wg21.link/p2183r0>

[P2195R0]

Bryce Adelstein Leibach. [Electronic Straw Polls](https://wg21.link/p2195r0). 15 September 2020. URL: <https://wg21.link/p2195r0>

[P2202R0]

Gašper Ažman, Tony van Eerd, Thomas Rodgers, Tomasz Kamiński, Corentin Jabot, Robert Leahy, Gordon Brown, Kirk Shoop, Eric Niebler, Dietmar Kühl. [Senders/Receivers group Executors review report](https://wg21.link/p2202r0). 15 July 2020. URL: <https://wg21.link/p2202r0>

[P2203R0]

Ryan McDougall, Tomasz Kamiński, Conor Hoekstra, Bryan St. Amour. [LEWG Executors Customization Point Report](https://wg21.link/p2203r0). 15 July 2020. URL: <https://wg21.link/p2203r0>

[P2205R0]

Inbal Levi, Ruslan Arutyunyan, Zach Laine, Tom Scogland, Dmitri Mokhov, Chris Kohlhoff, Daisy Hollman, Jared Hoberock. [Executors Review - Polymorphic Executor](https://wg21.link/p2205r0). 19 August 2020. URL: <https://wg21.link/p2205r0>

[P2206R0]

Ruslan Arutyunyan, Ben Craig, Guy Davidson, Anuya Welling, Michael Voss, Tom Scogland, Chris Kohlhoff, Gordon Brown, Lee Howes. [Executors Thread Pool review report](https://wg21.link/p2206r0). 2 September 2020. URL: <https://wg21.link/p2206r0>

[P2207R0]

Guy Davidson, Ben Craig, Robert Leahy, Michał Dominiak, Alexey Kukanov, Hartmut Kaiser, Daisy Hollman, Jared Hoberock, Gordon Brown. [Executors review: concepts breakout group report](https://wg21.link/p2207r0). 14 August 2020. URL: <https://wg21.link/p2207r0>

[P2212R1]

Alexey Dmitriev, Howard Hinnant. [Relax Requirements for time_point::clock](https://wg21.link/p2212r1). 14 September 2020. URL: <https://wg21.link/p2212r1>

[P2235R0]

Ville Voutilainen. [Disentangling Schedulers and Executors](https://isocpp.org/files/papers/P2235R0.html). 2020-10-15. URL:
<https://isocpp.org/files/papers/P2235R0.html>